



BUTACU IONEL-CATALIN

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SUMMARY

ML Engineer and Software Engineer with 3+ years of professional experience building production-grade software at a commercial CAD/CAM company. Pursuing a Master's in Artificial Intelligence with hands-on end-to-end ML projects spanning computer vision, generative AI, NLP, and reinforcement learning.

PROFESSIONAL EXPERIENCE

Gemini CAD Systems

Software Engineer

Iași, RO

June 2023 – Present

- Built features spanning Qt's UI, networking, model/view, and event-handling layers
- Decomposed complex requirements into scoped deliverables, consistently meeting sprint estimations.
- Reduced technical debt through targeted refactoring and clean code, improving codebase resilience.
- Diagnosed and resolved production crashes and regressions through systematic root cause analysis.
- Participated in the full Scrum lifecycle: sprint planning, estimation, code reviews, sprint demos, and retrospectives.

Software Engineer Intern

August 2022 – May 2023

- Built UI components from design specifications, integrating them into existing production views.
- Delivered pixel-accurate UI and extended feature parameters through import/export integrations.
- Practised all Git workflow stages: branching, commits, merge requests, code reviews, and CI/CD.

Special Telecommunications Service (STS)

IT Support Staff — Electoral IT Systems

Botoșani, RO

August 2020 – September 2020

- Provided technical support for electoral IT systems (e.g., SIMPV/SICPV).
- Assisted polling-station staff with application usage and hardware operation.

CERTIFICATIONS

- Oracle Cloud Infrastructure 2025 Data Science Professional (1Z0-1110-25) [🔗](#)
- Oracle Cloud Infrastructure 2025 Generative AI Professional (1Z0-1127-25) [🔗](#)
- Oracle Cloud Infrastructure 2025 AI Vector Search Professional (1Z0-184-25) [🔗](#)
- Oracle Analytics Cloud 2025 Certified Professional (1Z0-1041-25) [🔗](#)

SKILLS & COMPETENCIES

- Deep Learning Architecture Design
- End-to-End ML Pipeline Development
- NLP & Large Language Model Engineering
- Feature Engineering & Data Preprocessing
- MLOps Pipeline Development
- Model Performance Optimisation & Evaluation
- Computer Vision System Design
- Production ML Deployment & Scaling

EDUCATION

Technical University of Iași

Master of Science in Artificial Intelligence

Iași, RO

October 2024 - June 2026

Advanced Coursework: Neural Networks, Machine Learning, Natural Language Processing (NLP), Reinforcement Learning, Multi-Agent Systems, Cloud Computing, Big Data Techniques, Edge Computing Applications, Intelligent Cyber-Security

Bachelor of Science in Computer Engineering

October 2020 - June 2024

Core Curriculum: Advanced Computer Programming, Data Structures & Algorithms, Real-Time Programming, Database Management Systems, Systems Theory, Microprocessor-Based Systems, Embedded Systems Engineering, Control Engineering, Computer Networks, Neural Networks & Fuzzy Logic, Knowledge-Based Systems

PROJECTS

GenAI with Physics Constraints (in progress) 	2026
Eight-module pipeline converting natural-language prompts into physics-realistic MP4 videos. Parses text into a 3-D scene, generates assets (Shap-E), synthesises motion (SSM/KIT-ML dataset), simulates rigid-body physics (PyBullet), and optionally enhances frames with ControlNet.	
IDS/IPS Smart Firewall 	2026
Team project (3 members); personal contribution: DBSCAN-based smart firewall that clusters observed attack payloads at runtime and auto-generates Snort rules, reducing ML detector load. Full system also includes a CNN classifier (95%+ accuracy) and an RL+BiLSTM adversarial attack agent.	
Hand Gesture Recognition 	2025
End-to-end YOLOv5n object-detection pipeline for hand gesture recognition (HaGRID dataset). Optimised for NPU/CPU deployment via TFLite INT8 quantisation, CLAHE contrast preprocessing, and temporal smoothing to reduce false positives.	
AI Medication Assistant 	2025
Full-stack RAG application: React/TypeScript frontend (GitHub Pages) backed by a Gradio API (Hugging Face Spaces). FAISS-indexed knowledge base of 1200+ medications across 20 medical categories; answers symptom queries and surfaces direct pharmacy links.	
Metaheuristic Framework 	2025
Python library implementing Canonical GA, Real-Valued GA, Differential Evolution, and Random Search. Each solver is benchmarked on standard mathematical test functions with automated CSV reporting and convergence visualisation.	
fMRI Brain-Age Prediction 	2025
Competition ML pipeline predicting biological brain age from ~20k fMRI neuroimaging connectivity features. Gender-stratified ElasticNet model with spectral feature selection; ranked 8th nationally at the WiDS Datathon 2025 University Edition.	

VOLUNTEER EXPERIENCE

LOCAL BEST Group Iași	Iași, RO
IT Department Coordinator	Sep 2023 – Aug 2024
- Directed IT operations and maintained technical infrastructure across organizational events. - Mentored volunteers and oversaw delivery of technical projects. - Administered hardware, software licensing, and network resources.	
Event Coordinator JobShop Career Event	Nov 2022 – Apr 2023
- Coordinated logistics and operations for a large student–industry career event. - Managed scheduling, partner engagement, and on-site execution. - Improved attendee experience through strategic planning and workflow optimization.	
TOPICS Responsible Prove It 24H Techathon	Dec 2021 – Mar 2022
- Defined technical requirements and coordinated hackathon logistics. - Supported participants with rules, resources, and real-time assistance. - Facilitated communication between organizers, industry mentors, and teams.	
IT Responsible BEST Training Weekend	Jan 2021 – May 2022
- Developed and maintained the event website using HTML, CSS, and JavaScript. - Collaborated with design and marketing responsables on content, branding, and technical setup.	

ADDITIONAL SKILLS

ML / Frameworks:	PyTorch, TensorFlow, Scikit-learn, Hugging Face, LangChain&LangGraph, OpenCV
MLOps & Cloud:	Docker, MLflow, W&B, ONNX, Oracle OCI, Azure ML
Data & Dev:	Python, C++, SQL, Spark, Flask, FastAPI, React, NodeJs
Soft Skills:	Problem Decomposition, Clear Communication, Knowledge Sharing, Self-directed Learning